

TIME TO REMOVE THE TRASH

From Space Around Earth

Consequent to increase in number of artificial objects orbiting Earth, collision hazards have become more probable. Spacecraft enabling communication, navigation, scientific, and other missions now share Earth orbit with spent rocket bodies, nonfunctional spacecraft, fragments from spacecraft breakups, and other debris created as a byproduct of space operations. Millions of tiny pieces of junk orbiting the Earth could have a major impact on space. Orbital debris removal has become a very critical part of the commercial and scientific space management. It is an aggregating risk which needs to be addressed to prevent loss of spacecraft to debris collision. This article aims to create awareness about this risk. Innovative techniques have been proposed to solve the problem



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From almost the first moment that man started traveling beyond Earth's atmosphere, all sorts of debris in space has been left. Not only is it wasteful, space junk can be dangerous to satellites, to space stations, and when some of it plummets back to Earth, to human life on the ground.

Two years ago, an 18-ton Chinese rocket plummeted toward Earth and landed in the Atlantic Ocean, becoming one of the largest pieces of space debris to fall uncontrolled back to Earth in recent times. The *Times of India* in their issue dated 11th April 2022 reported that India alone has 103 active or defunct spacecraft and 114 objects categorised as space debris.

Such space debris is composed of spent boost stages, collision fragments, human discards, inactive or defunct

satellites, rocket bodies, ISS construction material, and other parts of spacecraft that have been left behind.

Some space junk results from collisions or anti-satellite tests in orbit. When two satellites collide, they can smash apart into thousands of new pieces, creating lots of new debris. This is rare, but several countries including the US, China, and India have used missiles to practice blowing up their own satellites.

There are estimated to be over 128 million pieces of debris smaller than one centimetre. There are approximately 900,000 pieces from one to ten centimetre in size. The current count of large debris (10cm across or larger) is 34,000.

Space debris is mostly concentrated in the near-Earth space region, in the Low Earth Orbit (LEO) and Geostationary Earth Orbit (GEO) regions. Low Earth Orbit is defined as the region of space around Earth within an altitude of 160 to 2000km wherein a large number of active satellites operate. This causes a substantial operational risk, ranging from the need to perform evasive maneuvers to defects or even obliteration of spacecraft due to collisions with pieces of debris, which at orbital speeds of approximately 7.5km/s can cause considerable damage.

NASA estimates that currently there are some 21,000 pieces of space junk larger than a softball orbiting the Earth

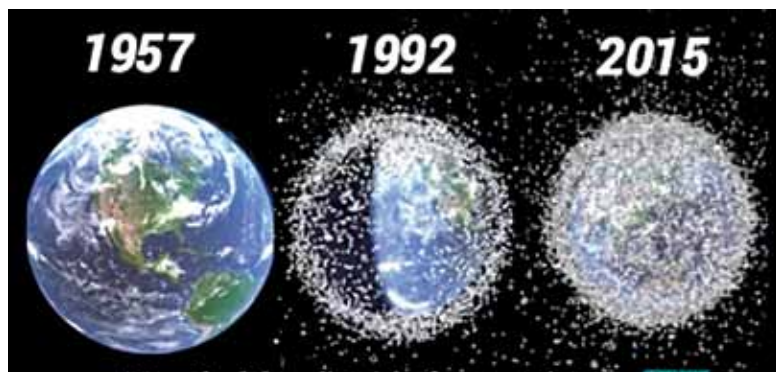


Fig. 1: Since humans first went up in space in 1961, the amount of space junk around Earth has hit a critical point